CAMBRIDGE SPARK

Product Management in AI

www.cambridgespark.com

Agenda

1

Introduction and Feedback on Pre-workshop Activities

2

Product LifeCycle and Creation Process

3

Business Model and Testing

4

User Journey Mapping and Machine Learning Canvas

Learning Outcomes

- Solid understanding of Product Management
- Differences between Product Manager and Product Manager in AI
- Learning about two canvas which will help you in your PM role
- Introduce to the learners the techniques and methodologies used by Product Managers
- Cover the different KSBs highlighted for the course

Knowledge, Skills and Behaviours

This module will help you to develop the following KSBs for your apprenticeship:

- **K5:** How to design and deploy effective techniques of data analysis and research to meet the needs of the business and customers.
- **K6:** How data products can be delivered to engage the customer, organise information or solve a business problem using a range of methodologies including iterative and incremental development and project management approaches.
- **K7:** How to solve problems and evaluate software solutions via analysis of test data and results from research, feasibility, acceptance and usability testing.

Knowledge, Skills and Behaviours

- **K14:** The business value of a data product that can deliver the solution in line with business needs, quality standards and timescales
- **K15:** The engineering principles used (general and software) to investigate and manage the design, development and deployment of new data products within the business
- **K17:** How to identify current industry trends across AI and data science and how to apply these

Knowledge, Skills and Behaviours

This module will help you to develop the following KSBs for your apprenticeship:

- **S5:** Manage expectations and present user research insight, proposed solutions and/or test findings to clients and stakeholders.
- **S6:** Provide direction and technical guidance for the business with regard to AI and data science opportunities.
- **S8:** Coordinate, negotiate with and manage expectations of diverse stakeholders suppliers with conflicting priorities, interests and timescales.

Knowledge, Skills and Behaviours

- **S24:** Apply research methodology and project management techniques appropriate to the organisation and products
- **S28:** Undertakes independent, impartial decision-making respecting the opinions and views of others in complex, unpredictable and changing circumstances

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Before we begin - Miro Refresher

- We are going to share our work on Miro
- Miro is a digital tool useful for workshops and sharing information
- Move around with the cursor or trackpad
- At any time on the top right corner, you can hide cursors from other participants or come back to my view by clicking on my “head”
- You can add sticky notes by pressing N
- That’s all you need to know for now!

Exercise 1

Ice Breakers

Exercise 2

Choose your workshop

Reflect on the pre-workshop activities and answer the following questions:

- **What would you like to go deeper into in the Product Manager role?**
- **What was unclear about the key trends section?**
- **What questions do you have to compare to your current role?**
- **What you feel you lacked in the pre-workshop activities?**

Product Management

What is a GOOD product manager?

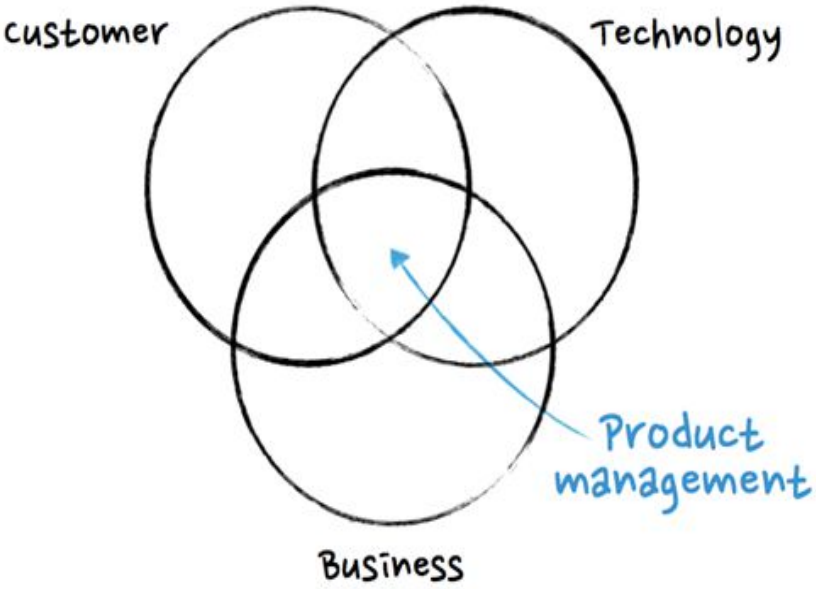
- Ensure that companies build the right thing, at the right time, the right way.
- Gather knowledge around the product.
- Analyse and find the best solution that suits the company goals.
- Learn how to compromise, negotiate and deal with multiple requests all having a critical priority.

You **manage the product** AND you need to **use influence and collaborate** through relationships

What is a GOOD product manager?

“CEO of the product”

- Ben Horowitz



Product VS Project Management

PRODUCT MANAGEMENT

- Strategic Role
- Continuous delivery and evolution of a product
- Consumer and Market Research
- Identifying needs, pains and opportunities
- Creating a vision
- Building a roadmap

PROJECT MANAGEMENT

- Tactical Role
- Implementation of the project on time and budget
- Project Planning
- Resource Scheduling
- Risk Management

Terminology in Product Management

- KPI
- Agile, Stories, Story Points, Scrum
- UX-UI
- Persona
- Customer Interviews
- Prototyping
- Low, High Fidelity
- Go To Market
- Product Market Fit
- Mockups
- Roadmap
- Usability
- A/B testing
- Rollout

Open Discussion

What is the place of the product manager in a digital organisation?

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Understanding Product

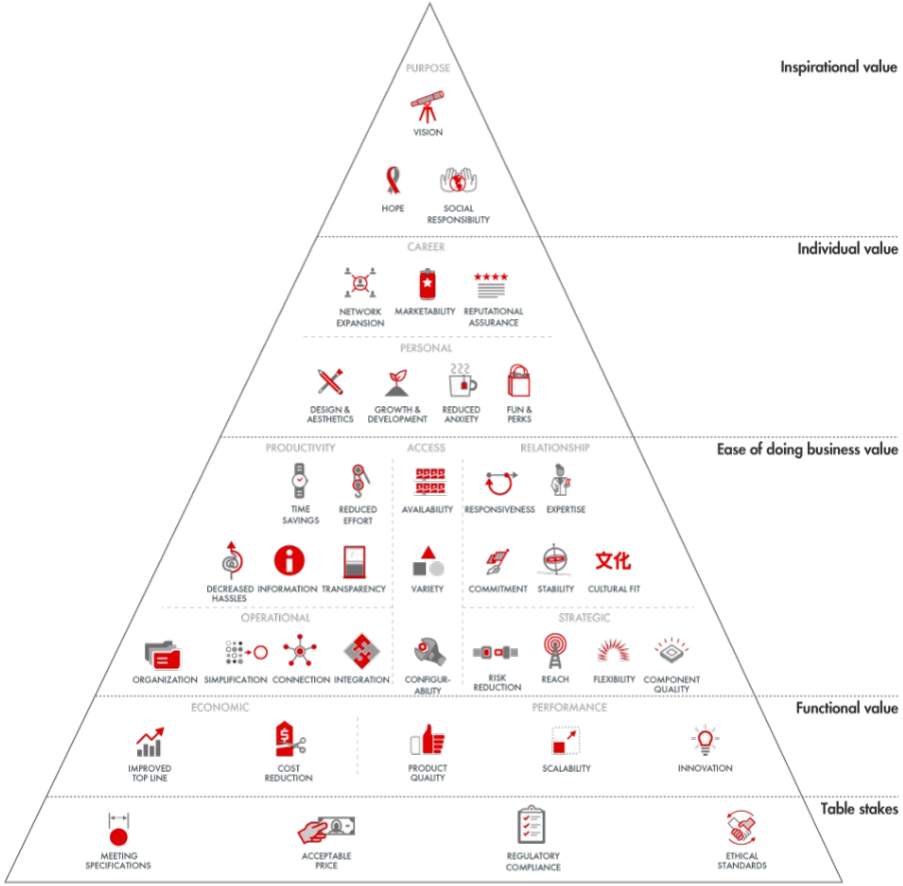
90%

of SaaS startups fail, only 35% get past the 10-year mark, and only 40% of these ever become profitable.

Some reasons why startups fails

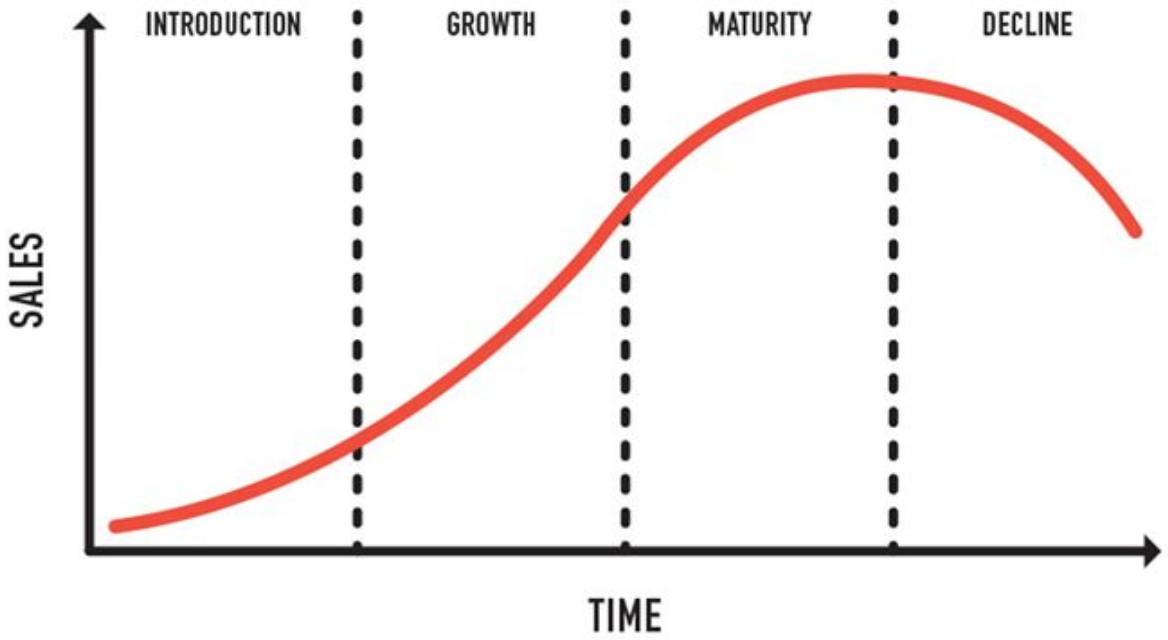
- No product-market fit
- Bad positioning
- Struggle to identify user problems
- MVP doesn’t do the job it intended to do
- No time spent on market research and customer discovery
- Poor user experience
- Users can’t find the value quickly in the product
- Go-To-Market strategy is overlooked
- Price strategy can be complex
- Onboarding is complicated and cumbersome
- Rely on an old sales model

What people are looking into a product?

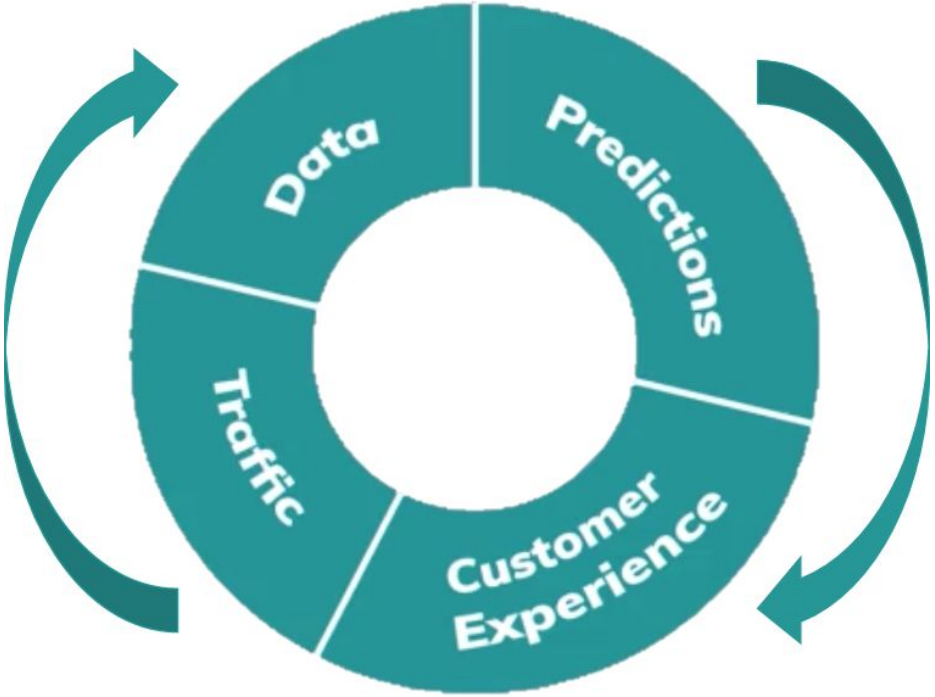


Product Lifecycle

Traditional Product Life Cycle



New Product Life Cycle in Digital Organisation



What is the process followed by PMs?

1. Conduct research and develop user understanding
2. Identify patterns and concepts
3. Generate ideas
4. Build prototype and validate hypotheses
5. Implement and Go-To-Market

Creating a new product

Research Exercise

As a team, I want you to choose one of the following idea and start creating a new product.

Think about it in regards to AI.

- a service such as a taxi, phone company, or laundry service
- a retail business such as a coffee shop or hot dog stand
- improving an app such as Google, CityMapper, Netflix, Instagram
- upgrade one department at their work onboarding, customer success, sales, marketing, or HR

The first step in creating a new product or service is the research and ideation phase.

Using the following methodology, as a team, begin your first step in creating a brand new product.

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Business Model

Business Model Definition

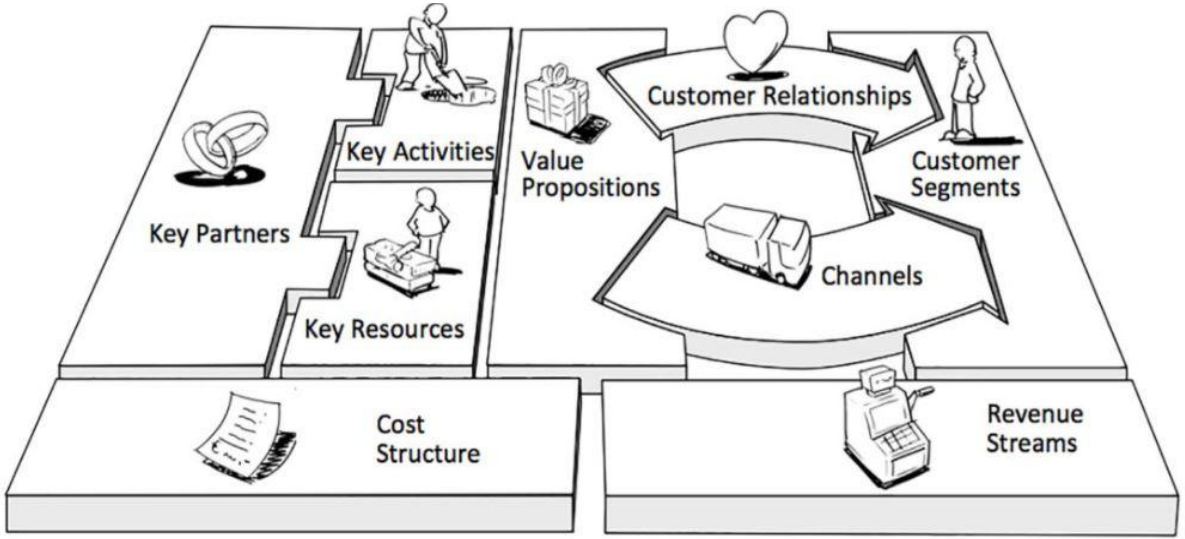
- The first and only reason for a business to exist is to answer a need, to solve a problem.
- If there is no problem to address, there is no business to make!

A Business Model is a set of verbalized assumptions about how a company or product :

- Creates and delivers value and,
- Reaps the profits of delivering this value.

Business Model Canvas

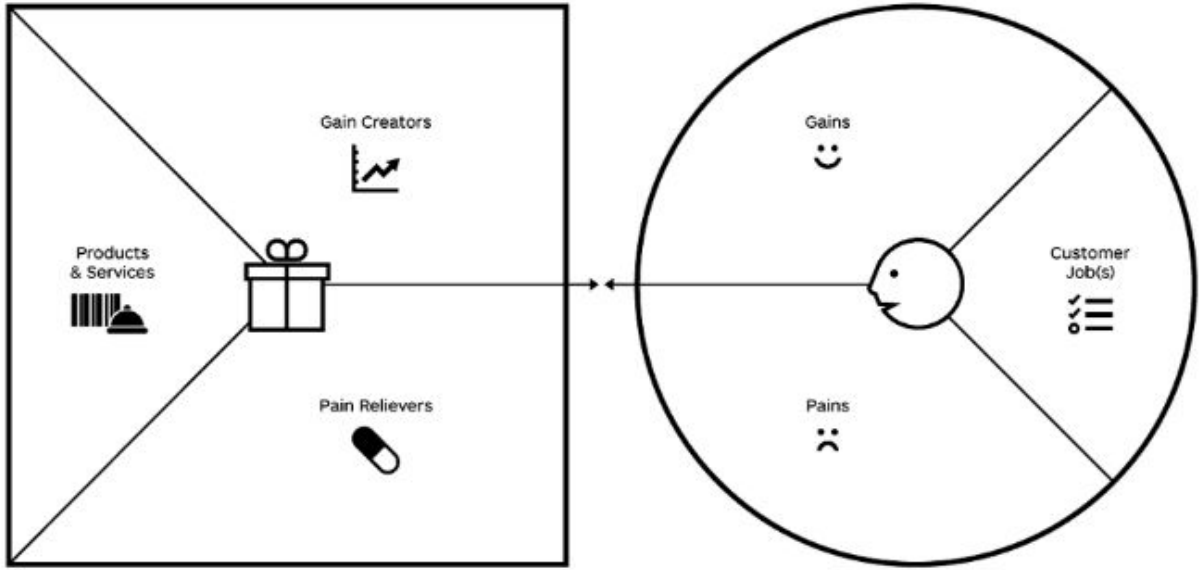
- It's **a tool for visualising a Business Model.**
- It consists of **interlinked key elements** that are common for most business models.



Persona And Value Proposition

Value Proposition

- Reasons why customers turn to your company
- A value proposition solves a customer's problem and cover requirements for one specific Customer Segment
- ***It's better to be a pain reliever than a vitamin!***



Business Model Canvas Example

Key Partners

- Customer Service
- Financial Services
- Legal

Key Activities

- Machine Learning model
- Financial knowledge and advices

Key Resources

- AI assistant and recommendation system

Value Proposition

- Automatically sell new services to clients through an AI financial assistant

Customer Relationships

- Personal Assistance

Channels

- Mobile App/ Messaging

Customer Segments

- Young professionals with some investment and cash on hands

Cost Structure

- Machine Learning Data
- Channel Costs

Revenue Streams

- New services fees
- Subscription for premium recommendations

Learnings

Business Model Exercise

- Identify a Value Proposition on your new product or service.
- Design a business model based on the Value Proposition created.

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User Journey Mapping

User Journey

- User Journey is a **flow of action a customer may take when using a product or service.**
- User Journey Map is a **visualisation tool** a customer goes through to accomplish a goal.

User Journey

- **Strangers** - They don’t know your service or brand
- **Prospects** - First encounter with your product - customers start to engage with your brand
- **Customers** - Customers are engaged and find value
- **Advocates** - Customers are compelled by your product and refer it to their friends and family

Open Discussion

Why is the User Journey so important in any role relating to product creation?

Machine Learning Canvas

What is a Machine Learning Canvas?

- It’s a tool for visualising the learnings elements happening in the intelligent system
- It consists of interlinked key elements from the value proposition, the learning elements to the predictions
- The What + Why + Who and the How

Machine Learning vs Business Model

Business Model

- Strategic Thinking
- Linking stakeholders with the product management and dev team
- What is the value proposition?
- Which consumers are we serving?
- How are we serving them?

Machine Learning

- Execution Thinking
- Linking with the team together: PM, Designer, Data Scientist, Data Engineers
- What data are we learning from?
- What kind of predictions we are making and how are we using them?
- A snapshot that can evolve through time

The Machine Learning Canvas (v0.4)

Designed for: Designed by: Date: Iteration:

Decisions

How are predictions used to make decisions that provide the proposed value to the end-user?

Each month we randomly filter out 50% of the clients
Sort remaining by descending and show prediction path on recommendations
Send recommendation as many as suggested by simulation

ML task

Input, output to predict, type of problem.

Predict answer "Is this client going to take our recommendation?"
- Input Customer
- Output: "Took" or "Didn't' take"
- Binary classification

Value Propositions

What are we trying to do for the end-user(s) of the predictive system? What objectives are we serving?

We want our clients to get new opportunities automatically based on recommendations based on their profile and their portfolios and interests
Improve automatically subscription to financial services

Data Sources

Which raw data sources can we use (internal and external)?

- CRM tool
- Customer Support
- Financial Services
- Customer data

Collecting Data

How do we get new data to learn from (inputs and outputs)?

Every month, we see which of last month's customers took the recommendations by looking in the database

Making Predictions

When do we make predictions on new inputs? How long do we have to featurize a new input and make a prediction?

Every month we featurize all current customers and make predictions for them

We do this overnight along with building the model that powers these predictions and evaluating it

Offline Evaluation

Methods and metrics to evaluate the system before deployment.

Before soliciting customers
- Evaluate new model accuracy on predefined profiles
- Simulate recommendation taken in the last month (using model from customers 2 month ago)

Features

Input representations extracted from raw data sources.

- Consumer info at time (age, sex, city, job...)

Event between t-1 month and t
- Recommandations taken
- Support or financial services interactions

Building Models

When do we create/update models with new training data? How long do we have to featurize training inputs and create a model?

Every month we create a new model from the previous month data set

We do this overnight with offline evaluation and making predictions

Live Evaluation and Monitoring

Methods and metrics to evaluate the system after deployment, and to quantify value creation.

- Accuracy of last month prediction on hold out set
- Compare recommandation taken and earn revenue between last month hold out set and remaining set
- Monitor (non recommended customer/ # solicitations)

machinelearningcanvas.com by Louis Dorard, Ph.D.

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Learnings

Machine Learning Canvas Exercise

- Based on the user journey and business model, identify the Machine Learning Canvas for your idea

Learning Outcomes

- Solid understanding of Product Management
- Differences between Product Manager and Product Manager in AI
- Learning about two canvas which will help you in your PM role
- Introduce to the learners the techniques and methodologies used by Product Managers
- Cover the different KSB highlighted for the course

What's next

- Log onto EDUKATE.AI and complete your post-workshop activities.
- Review the KSBs covered in this module and think about how you could use these in your role.
- Drop into an office hours session to discuss how to apply what you have learned to your work-based learning activities.

Thank You

Send a message or connect with us through social media, to find out more about how you can learn more

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